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Roll No

ME-802 (GS)
B.E. VIII Semester Examination, June 2020
Grading System (GS)
Machine Design
Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. A pair of bevel gears, with 20° pressure angle, consists of a 20 teeth pinion meshing with a 30 teeth gear. The module is 4mm, while the face width is 20mm. The material for the pinion and gear is steel 50C4 ($S_{ut} = 750 \text{ N/mm}^2$). The gear teeth are lapped and ground (Class - 3) and the surface hardness is 400 BHN. The pinion rotates at 500 rpm and receives 2.5 kW power from the electric motor. The starting torque of the motor is 150% of the rated torque. Determine the factor of safety against bending failure and against pitting failure.

2. Compare belt drive, chain drive and rope drive for power transmission.

OR

Derive an expression for power rating of a V-belt drive.

3. What are the advantages of cycloidal teeth gears and involute teeth gear?

4. Following are the details of a crossed belt drive

Diameter of the driver	:	200mm
Diameter of the follower	:	400mm
Center distance of the drive	:	2m
Speed of the drive	:	400rpm
Angle of contact	:	197.3°

Determine the length of the belt required.

5. Find the maxima and minima, if any of the functions

i) $f(x) = 4x^3 - 18x^2 + 27x - 7$

ii) $f(x) = x^4 / (x-1)(x-3)^3$

6. What do you mean by slip in a belt drive? Differentiate between open belt and crossed belt drive. List out the applications of belt drives.

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7. Why are two universal joints often used when there is angular misalignment between two shafts?
8. Answer any two of the following:
 - a) What do you understand by simplex, duplex and triplex chains?
 - b) State the function of the following in an internal combustion engine:
 - i) Ribs
 - ii) Piston Skirt
 - iii) Piston Rings
 - c) Compare the stress distribution in a thin and thick walled pressure vessel.
